CODECRAFT 26

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Emergency Rescue Request Prioritization and Management System

1. Introduction

This project is developed using C programming to manage emergency rescue requests. It helps to store request details, display critical cases first, assign rescue teams, update status, and search for requests.

2. Objective

The main objective is to manage emergency requests easily and make sure critical cases are given more importance than normal cases.

3. Technologies Used

- Programming Language: C
- Compiler: GCC
- Concepts: Structures, arrays, loops, and conditional statements.

4. Modules

1. Register Request: Add emergency details such as ID, location, people affected, emergency level, and resources.
2. Display Requests: Show critical requests first, followed by other requests.
3. Assign Team: Assign a rescue team to a request.
4. Update Status: Change the request status to Pending or Completed.
5. Search Request: Find a request using its ID.
6. Emergency Summary: Display total requests, critical cases, normal cases, completed requests, and pending requests.

7. Exit: Close the program.

5. Algorithm

1. Start the program.
2. Display the menu.
3. Get the user's choice.
4. Perform the selected operation.
5. Display the result.
6. Repeat until the user selects Exit.
7. Stop the program.

6. Advantages

- Makes emergency details easy to manage.
- Displays critical cases first.
- Helps assign rescue teams.
- Makes searching and status updating easier.
- Provides a summary of emergency requests.

7. Conclusion

This project helps manage emergency requests in a simple way using C programming. It allows users to register requests, identify critical cases, assign teams, update statuses, and view emergency summaries.

PROGRAM:

```c
#include <stdio.h>
#include <string.h>
struct Request {
    char id[10];
    char location[30];
    int people;
    char emergency[20];
    char resources[30];
    char team[30];
    char status[20];
};
struct Request list[10];
int total_requests = 0;

int main()
{
    int choice;
    char search_id[10];
    int found;
    int critical_count, normal_count, completed_count, pending_count;
    while(1)
    {
        printf("Enter choice (1-7): ");
        scanf("%d", &choice);
        if (choice == 1)
        {
            if (total_requests >= 10) {
                printf("System full! Cannot add more requests.\n");
            }
            else {
                printf("\n  REGISTRATION: \n");
                printf("Request ID: ");
                scanf("%9s", list[total_requests].id);
                printf("Location: ");
                scanf("%29s", list[total_requests].location);
                printf("People: ");
```

```
                scanf("%d", &list[total_requests].people);
                printf("Emergency Level:");
                scanf("%19s", list[total_requests].emergency);
                printf("Resources: ");
                scanf("%29s", list[total_requests].resources);
                strcpy(list[total_requests].team, "None");
                strcpy(list[total_requests].status, "Pending");
                printf("Successfully Registered!\n");
                total_requests++;
            }
        }
        else if (choice == 2)
        {
            if (total_requests == 0) {
                printf("No requests found.\n");
            }
            else {
                printf("\n--- CURRENT REQUESTS ---\n");
                for (int i = 0; i < total_requests; i++) {
                    if (strcmp(list[i].emergency, "Critical") == 0) {
                        printf("[%s] Loc: %s -People: %d -Level: %s -Team:
%s -Status: %s\n",
                               list[i].id, list[i].location,
list[i].people,
                               list[i].emergency, list[i].team,
list[i].status);
                    }
                }
                for (int i = 0; i < total_requests; i++) {
                    if (strcmp(list[i].emergency, "Critical") != 0) {
                        printf("[%s] Loc: %s -People: %d Level: %s | Team:
%s | Status: %s\n",
                               list[i].id, list[i].location,
list[i].people,
                               list[i].emergency, list[i].team,
list[i].status);
                    }
                }
            }
        }
```

```
        else if (choice == 3)
        {
            printf("Enter Request ID to assign team: ");
            scanf("%9s", search_id);
            found = 0;
            for (int i = 0; i < total_requests; i++) {
                if (strcmp(list[i].id, search_id) == 0) {
                    printf("Enter Rescue Team Name: ");
                    scanf("%29s", list[i].team);
                    printf("Team Assigned!\n");
                    found = 1;
                    break;
                }
            }
            if (found == 0)
             printf("Request ID not found.\n");
        }
        else if (choice == 4)
        {
            printf("Enter Request ID to update status: ");
            scanf("%9s", search_id);
            found = 0;
            for (int i = 0; i < total_requests; i++) {
                if (strcmp(list[i].id, search_id) == 0) {
                    printf("Enter New Status (Pending/Completed): ");
                    scanf("%19s", list[i].status);
                    printf("Status Updated!\n");
                    found = 1;
                    break;
                }
            }
            if (found == 0)
            printf("Request ID not found.\n");
        }
        else if (choice == 5)
        {
            printf("Enter Request ID to search: ");
            scanf("%9s", search_id);
            found = 0;
            for (int i = 0; i < total_requests; i++) {
```

```
                if (strcmp(list[i].id, search_id) == 0) {
                    printf("\n--- DETAILS FOUND ---\n");
                    printf("ID: %s\nLocation: %s\nPeople: %d\nLevel:
%s\nResources: %s\nTeam: %s\nStatus: %s\n",
                           list[i].id, list[i].location, list[i].people,
                           list[i].emergency, list[i].resources,
list[i].team, list[i].status);
                    found = 1;
                    break;
                }
            }
            if (found == 0)
            printf("Request ID not found.\n");
        }
        else if (choice == 6)
        {
            critical_count = 0; normal_count = 0; completed_count = 0;
pending_count = 0;

            for (int i = 0; i < total_requests; i++) {
                if (strcmp(list[i].emergency, "Critical") == 0)
critical_count++;
                else normal_count++;

                if (strcmp(list[i].status, "Completed") == 0)
completed_count++;
                else pending_count++;
            }
            printf("SUMMARY \n");
            printf("Total Requests: %d\n", total_requests);
            printf("Critical Cases: %d\n", critical_count);
            printf("Normal Cases:   %d\n", normal_count);
            printf("Completed:      %d\n", completed_count);
            printf("Pending:        %d\n", pending_count);
        }
        else if (choice == 7)
        {
            printf("Exiting Program...\n");
            break;
        }
```

```
        else
        {
            printf("Invalid selection! Use options 1 to 7.\n");
        }
    }
}
```

Output:

```
PS E:\C_CodeCraft26> gcc test.c -o test
PS E:\C_CodeCraft26> .\test.exe
1. Register Request
2. Display Requests (Critical First)
3. Assign Rescue Team
4. Update Status
5. Search Request
6. Generate Summary
7. Exit
Enter choice (1-7): 2
No requests found.
1. Register Request
2. Display Requests (Critical First)
3. Assign Rescue Team
4. Update Status
5. Search Request
6. Generate Summary
7. Exit
Enter choice (1-7): 2
No requests found.
1. Register Request
2. Display Requests (Critical First)
3. Assign Rescue Team
4. Update Status
5. Search Request
6. Generate Summary
7. Exit
Enter choice (1-7): 7
Exiting Program...
PS E:\C_CodeCraft26> gcc test.c -o test
PS E:\C_CodeCraft26> .\test.exe
Enter choice (1-7): 2
```

No requests found.
Enter choice (1-7): 5
Enter Request ID to search: yt
Request ID not found.